



Fig. 3: Image captured by Kinect after running `glview` command

freenect library that can take the data (such as colour, depth and other information) from Kinect and convert it into a format that is readable by OpenCV library for further processing and developing complex algorithms like face recognition.

Installing OpenCV. For installing OpenCV in Ubuntu 12.04, use the following commands:

```
$ sudo apt-get install build-essential libtiff4-dev cmake git libgtk2.0-dev python-dev python-numpy libavcodec-dev libavformat-dev libswscale-dev libstreamer0.10-dev libstreamer-plugins-base0.10-dev libv4l-dev libjpeg-dev libjasper-dev libtiff4-dev libdc1394-22-dev libxine-dev
```

```
$ cd ~
```

Download `OpenCv-2.4.7.tar.gz` from <http://sourceforge.net/projects/opencvlibrary/files/opencv-unix/2.4.7/opencv-2.4.7.tar.gz> download and extract it to home folder.

```
$ cd OpenCv-2.4.7/
$ mkdir build
$ cd build
$ cmake -D WITH_XINE=ON -D WITH_OPENGL=ON -D BUILD-EXAMPLE=ON
$ make
$ sudo make install
```

```
$ sudo gedit /etc/ld.so.conf.d/opencv.conf
```

At the end of the file, add the following lines and save it:

```
gedit /etc/ld.so.conf
/usr/local/lib
```

And then configure dynamic linker run-time bindings:

```
$ sudo ldconfig
```

OpenCV-2.4.7 has been installed. Open another terminal to check some of the samples:

```
$ cd ~
$ cd opencv-2.4.7/samples/c
$ chmod +x build_all.sh
$ ./build_all.sh
```

To check any sample, go to that folder and run the sample. For example, in the 'c' folder, we have contours, so to run it, enter the command given below:

```
$ cd ~
$ cd opencv-2.4.7/samples/c
$ ./contours
```

You will get the output image as shown in Fig. 2.

Installing libfreenect. To install libfreenect, we need to first install some prerequisites for building it:

```
$ sudo apt-get install git-core
cmake freeglut3-dev pkg-config build-essential libxmu-dev libxi-dev
```

```
libusb-1.0-0-dev
```

Now download the clone from 'github':

```
$ git clone git://github.com/
OpenKinect/libfreenect.git
```

Change directory to 'libfreenect'

```
$ cd libfreenect
```

Make a directory named 'build' and change current directory to 'build'

```
$ mkdir build
$ cd build
```

CMake is used to control the software compilation process with simple platform and compiler-independent configuration files. CMake generates native make files and workspaces that can be used in the compiler environment of your choice.

```
$ cmake ..
```

The make program uses the makefile database and latest modifications of the files to decide which of the files need to be updated.

Sudo means superuser do. It allows a system administrator to give certain users the ability to run some commands as root while logging all commands and arguments.

```
$ make
$ sudo make install
$ sudo ldconfig /usr/local/lib64/
```

Run the sample project:

```
$ sudo glview
```

Fig. 3 shows the image captured by the system after using `glview` command.

Now we are done with complete environment set-up and can program OpenCV using libfreenect. To write your own code for OpenCV, add the libfreenect.h header to your code. ●

The author is a FOSS advocate and his areas of interest include Kinect, Linux, Arduino and Image Processing using OpenCV. This article is based on his articles published earlier in EPY's Linux For You (now Open Source For You) magazine

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